Prodigy Anesthesia Quick Review

Calcium Channel Blockers

How do calcium channel blockers affect vascular smooth muscle? How does this affect a person's blood pressure?	Calcium channel blockers relax vascular smooth muscle, causing a reduction in blood pressure. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 203.*
Is orthostatic hypotension a common adverse side effect seen in patients who take calcium channel blockers?	No. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 203.*
In addition to vascular smooth muscle, what other smooth muscle is affected by the actions of calcium channel blockers?	Uterine, bronchiolar, and GI smooth muscle *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 203.*
List the calcium channel blockers that fall under the category of dihydropyridines.	Nifedipine, Nisoldipine, Nicardipine, Felodipine, Amlodipine, Isradipine *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 202.*
Once a calcium channel blocker binds to its receptor, how does the channel respond to depolarization in cardiac muscle? In smooth muscle?	The channel does not open as frequently, causing a drastic reduction in the transmembrane calcium current. This results in a reduction in the SA pacemaker rate, a reduction in the AV node conduction velocity, a decrease in contractility of cardiac muscle, and relaxation of smooth muscle. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 203.*
What are the 3 classifications of calcium channel blockers?	1. Dihydropyridines 2. Phenylalkylamines 2. Benzothiazepines *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 387.*
What specific calcium channel is predominant in smooth and cardiac muscle?	The L-type calcium channel *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 202.*
List the desirable effects of calcium channel blockers.	Decreased heart rate, a reduction in myocardial contractility, decreased activity of the SA node, reduced rate of cardiac impulse conduction through the AV node, relaxation of vascular smooth muscle resulting in vasodilation and a reduction in systemic blood pressure. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 388.*

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List common side effects of calcium-channel blockers.	Peripheral edema, headache, flushing, and systemic hypotension. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 387.*
What calcium channel blocker falls into the category of a benzothiazepine?	Diltiazem *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 388.*
What two calcium channel blockers possess local anesthetic activity?	Diltiazem and verapamil *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 395.*
What calcium-channel blockers are effective in the treatment of coronary artery spasm?	All calcium-channel blockers effectively treat coronary artery spasm. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 388.*
What two calcium-channel blockers exert the most negative inotropic effects?	Diltiazem and verapamil *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 388.*
What calcium channel blocker exerts the greatest degree of vasodilation?	Nicardipine *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 393.*
Verapamil should not be administered to patients with what conditions? Why?	Severe bradycardia, heart failure, dysfunction of the sinus node, or AV nodal block. Verapamil has a negative chronotropic effect on the SA node, a negative inotropic effect on cardiac muscle, and severely depresses the AV node. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 389.*
Why should verapamil be used cautiously in patients with Wolff-Parkinson-White syndrome?	Verapamil can augment conduction of cardiac impulses over accessory tracts through reflex sympathetic nervous system activity. This creates an increase in the rate of the ventricular response, which may lead to ventricular dysrhythmias. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 384, 389*

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What two drugs can be used to partially reverse toxicity caused by an overdose with a calcium channel blocker?	IV dopamine or calcium *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 394.*
What calcium channel blocker is useful in the prevention and treatment of cerebral vasospasm that often occurs with a subarachnoid hemorrhage?	Nimodipine *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 393.*
Do calcium channel blockers potentiate the effects of nondepolarizing and depolarizing neuromuscular-blocking drugs?	Yes. Calcium channel blockers do not have skeletal muscle relaxant properties on their own, but they can potentiate the effects of nondepolarizing and depolarizing neuromuscular-blocking drugs. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 395.*
What calcium channel blocker falls under the classification of a phenylalkylamine?	Verapamil *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 388.*

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